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The concentric circles model of the cultural industries

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This paper examines the assumptions and structure of the concentric circles model of the cultural industries. Empirical data for Australia, Canada, New Zealand, the UK and the US are used to illustrate the model's key characteristic: the proposition that the cultural content of the output of the cultural industries declines as one moves outwards from the core. The test uses the proportion of creative labour employed in production as a proxy for cultural content. The results confirm the model's validity as a means of depicting the structural characteristics of the cultural industries and also enable some wider features of the cultural workforce in the five countries to be examined.

Keywords: creative economy; cultural industries; concentric circles model; cultural occupations; cultural value; cultural policy

Introduction

The cultural industries have become an increasingly important focus of attention for cultural policy in a number of countries in recent years. Governments have begun to recognise the role of creativity as a key resource in driving innovation and promoting competitive advantage in a globalised world (Anheier & Isar, 2008). Concepts such as the creative economy have come into existence as a means of identifying the sectors of the macroeconomy producing creative goods and services in both developed and developing countries (Howkins, 2001; United Nations Conference on Trade & Development, 2008). At the heart of the creative economy are the cultural or creative industries, and hence a new orientation for cultural policy has begun to emerge, one that shifts the emphasis from a predominant concern with high culture and support for the arts to a more wide-ranging and pragmatic preoccupation with the cultural industries' contribution to employment creation and economic growth.¹ Of course the relationship between the cultural industries and cultural policy has been a long-standing matter for discussion within the cultural studies and political economy traditions (Garnham, 1990; Hartley, 2005; Hesmondhalgh & Pratt, 2005; Lewis & Miller, 2003; McGuigan, 2004; Pratt, 2005). However, the rapid technological change associated with globalisation that has occurred over the last decade or so has brought the cultural industries, and their role in cultural policy, into a new prominence.

But defining the cultural or creative industries remains a matter of some debate, and no clear agreement exists yet as to exactly which industries should or should not be included in the cultural sector of the economy. Furthermore, an essential requirement for effective policy formulation is a systematic understanding of the structure of the cultural economy and how its various parts fit together. There are a number of ways of interpreting the structural characteristics of the cultural industries, and indeed several different models have been put forward to describe and classify them, each with its own interpretation of the coverage of the cultural industries

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concept (Throsby, 2007). These models include: the well-known categorisation used in the development of the UK's creative industries policy strategy (Department of Culture, Media & Sport [DCMS], 2001); the copyright-based model adopted by the World Intellectual Property Organisation (WIPO, 2003); models derived from a cultural studies perspective in which these industries are seen as purveyors of symbolic texts (e.g. Hesmondhalgh, 2002); and a model developed within UNESCO's Institute for Statistics (2007) to provide a new classification system for the cultural industries as a basis for working towards international consistency in statistical collections.

All of these models differ in the composition of the list of industries that meet their criteria for inclusion, reflecting at least in part differences in the emphasis they place on an economic as distinct from a cultural interpretation of cultural goods and services. Thus, for example, the WIPO model focuses on the revenue-earning potential of intellectual property rights and hence adopts a clearly commercial orientation in its definition of cultural or creative commodities. Models originating in the field of cultural studies, by contrast, focus on the cultural value inherent in the products of the cultural industries, seeing this as the primary source of their economic worth (O'Connor, 2000, p. 5).

One model whose definition of cultural goods and services combines economic and cultural characteristics on more or less equal terms is the concentric circles model, in which the creative arts are placed at the centre and other industries are grouped around them. Seeing the arts as the core of the cultural industries has some precedents. For example, Garnham (1990, pp. 154–155) noted the focus of traditional arts policy on the central status of the creative artist, a focus that was decisively rejected by the Greater London Council's creative industries strategy in the 1980s. A pivotal role for the arts was also depicted in an unpublished report for the British Council (Gorham & Partners, 1987) in which the later stages in the artistic production chain (distribution, associated activities, etc.) were portrayed as concentric circles around the original creative arts. Subsequently Garnham (2001, p. 450) has pointed out that a central role for the artist has been reasserted in contemporary policy statements through reference to a "creative core", alongside a rejection of any distinction between high and popular culture.

As will be seen below, the model discussed in the present paper differs somewhat from these precursors. It was enunciated in general terms in Throsby (2001a, pp. 112–113) and subsequently developed in other places, for example in recent studies prepared for the European Commission (KEA European Affairs, 2006, pp. 53–57) and in the UK (The Work Foundation, 2007), where the concentric circles idea is used as a basis for classifying the creative industries in Europe and the UK respectively.²

This paper outlines the structure of the concentric circles model, considers the assumptions on which it is based, and derives some empirical evidence to illustrate the model's application by assembling data relating to the cultural industries in five countries (Australia, Canada, New Zealand, the UK and the US). The paper also examines some wider features of the employment contribution of the cultural industries in these countries.

The model

The concentric circles model is based initially on the proposition that cultural goods and services give rise to two distinguishable types of value: economic and cultural. Such a proposition can be used in principle as a basis for defining cultural goods and services as a distinct commodity class,³ and indeed is a distinction that is common to a variety of fields concerned with the value of cultural phenomena.⁴ The model asserts that it is the cultural value, or cultural content, of the goods and services produced that gives the cultural industries their most distinguishing characteristic.

Different goods have different degrees of cultural content relative to their commercial value; the model proposes that the more pronounced the cultural content of a particular good or service, the stronger is the claim of the industry producing it to be counted as a cultural industry. Thus are the concentric circles delineated: at the centre are core industries whose proportion of cultural to commercial content is judged according to given criteria to be highest, with layers extending outwards from the centre as the cultural content falls relative to the commercial value of the commodities or services produced.

How and by whom are these criteria to be determined? There are undoubtedly different interpretations as to how the cultural content of the output of a given cultural industry might be assessed. A student of cultural studies, for example, might look to the power and reach of the symbolic messages conveyed by given cultural products as an indicator of their cultural content, whereas a lawyer might use a scale related to the intellectual property rights involved. The concentric circles model adopts an assumption that cultural content springs from the incorporation of creative ideas into the production and/or presentation of sound, text and image and that these ideas originate in the arenas of primary artistic creativity. This is an assumption that accords primacy to the processes of artistic (as distinct from scientific) creativity, and is the reason why the creative arts – music, drama, dance, visual art, literature – lie at the centre of the model, with successive layers of the concentric circles defined as the ideas and influences of these creative activities diffuse outwards.

On this basis four layers or circles can be proposed to classify the industries that produce cultural goods and services according to the definitions discussed above. The categories and the main industries they contain are listed below and are illustrated in diagrammatic form in Figure 1.

Core creative arts

- Literature
- Music
- Performing arts
- Visual arts

Other core cultural industries

- Film
- Museums, galleries, libraries
- Photography

Wider cultural industries

- Heritage services
- Publishing and print media
- Sound recording
- Television and radio
- Video and computer games

Related industries

- Advertising
- Architecture
- Design
- Fashion

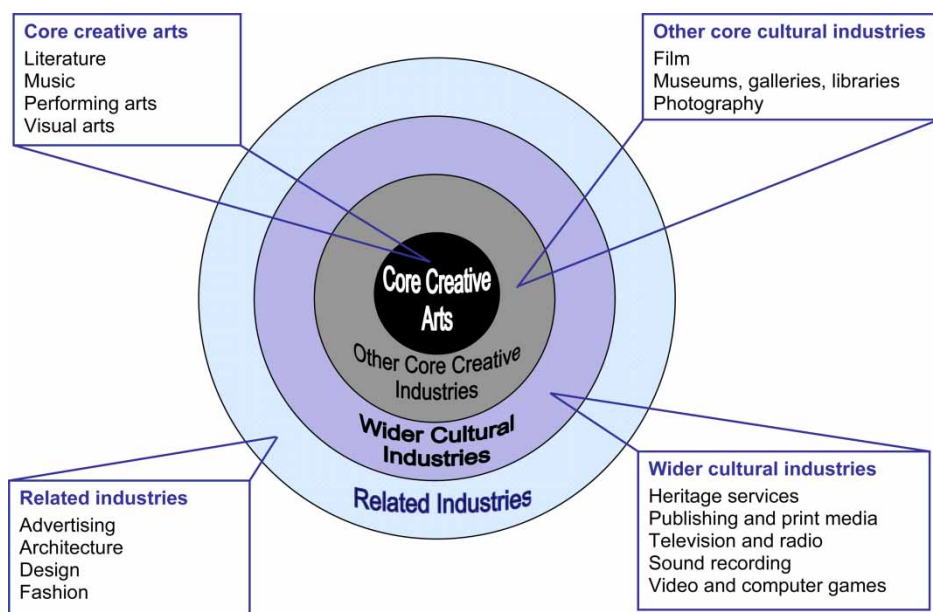


Figure 1. The concentric circles model of the cultural industries.

Although the above designation of four industry groups, and the allocation of particular industries between them, is derived from the set of assumptions on which the concentric circles model is based, decisions as to which industry goes where are essentially ad hoc;⁵ they do not rely on any objective benchmarks for assessing the cultural or commercial content of the goods and services produced. This is hardly surprising, since complex and multifaceted concepts such as cultural content, cultural value and creative ideas have no obvious metric. Nevertheless some *ex post* validation of the concentric circles model's classification system can be made by reference to empirical data, as discussed further below.

It should be noted that the "diffusion of creative ideas and influences" may occur through the sorts of generalised communication and exchange processes that govern the circulation of knowledge and information in the economy and society at large; for example, the plot of a novel or play may suggest ideas for a video or computer game. More concretely, the diffusion of ideas may arise through the fact that creative people who generate them actually work in different industries, providing direct input to the production of cultural content in industries further from the core; for example, a visual artist may have a creative practice producing original artworks, but may also work in the design industry, or an actor may appear on stage in the live theatre, as well as making television commercials in the advertising industry. However it happens, it is the creative ideas that generate the cultural content in the output of these industries.

As an interpretation of the structure of cultural production, the concentric circles model can be seen as a static snapshot at a given point in time, in contrast to a dynamic form of analysis such as a value chain model. In the latter, an original creative idea is traced from its origins through a value-adding process of production, distribution and marketing to final consumption. In the concentric circles model, downstream functions such as distribution are represented as distinct industries in their own right, incorporating original creative ideas produced in the core into their

production processes as intermediate inputs. For example, television scriptwriters, located at the core of the model, sell their work to broadcasters located in the “wider cultural industries” circle. At a given point in time the output of both industries – the scriptwriting industry and the television industry – can be observed and, under appropriate assumptions, the cultural content of their output assessed, as described in the next section.⁶

The need for empirical evidence

It is one thing to propose a classification of the cultural industries based on the relative level of cultural content in the industries’ various outputs; it is quite another to put the classification into effect on the basis of objective measurement. As noted above, cultural content has no immediately obvious unit of account, and in any case it is a concept where interpretation will vary according to the standpoint of the observer. Thus trying to assess cultural content from the output side would seem to offer little chance of success. However, it may be possible to obtain a proxy for cultural content by looking at the inputs used in cultural production.

Specifically, we might assume that the cultural content of output as defined in the concentric circles model is proportional to the level of creative input, and that the sole indicator of the input of creativity is the amount of creative labour used in production. Inputs of creative labour in various industries are likely to be measurable, for example if we can associate creative labour with what can be termed creative occupations. If occupations can be classified as creative or non-creative, it should be possible, using data on employment cross-tabulated by occupation and industry, to calculate for a given geographical entity such as a city, a state, a region, a country or a group of countries, the proportion of creative to total employment in each of the industries in the model. These proportions might then be taken as a proxy for the cultural content of the output of those industries according to the assumptions of the model. The hypothesis to be tested is that these proportions will decline as one moves outwards through the concentric circles of the model.

Such a procedure requires an initial assignment of occupations to creative or non-creative categories. In most areas of cultural production it is relatively easy to distinguish between creative and non-creative occupations: in the theatre industry, for example, actors, writers and directors are clear examples of creative occupations, whereas stagehands, ticket sellers and accountants can be labelled non-creative.⁷ Nevertheless the distinction cannot be regarded as entirely clear cut and unambiguous, and borderline cases will always arise. For example, how should one classify a writer such as a journalist, or a craftsperson making production runs of pottery items? Moreover some cultural outputs – for example in theatre, television and film – are produced by teams, where the creative input is diffused across all members of the group including those whose occupations may not be obviously creative in nature.

Despite these difficulties, a broadly workable classification should be possible, especially if a more precise definition of a creative occupation can be articulated. Such a definition might recognise, for example, three specific types of creative activity within the cultural sector (Throsby, 2003, p. 175): the production of primary creative output by occupations such as writers, composers, visual artists, film and video makers, sculptors, craftspeople, etc.; creative interpretation as practised by performers in dance, drama, music, etc. in a variety of media; and the supply of creative services in support of arts and cultural production by workers such as book editors, lighting designers, music producers and so on.⁸

It should be noted that if census data are to be used as the empirical source of employment numbers, the well-known deficiencies of such data in measuring the cultural workforce must be borne in mind.⁹ These problems include the question of whether or not census data can

adequately capture the numbers of practising professional artists. A difficulty in this respect arises because a worker's job for the purpose of the statistical collection is generally specified as the "main job" held in the week of the census; as a result a number of professional artists, who out of financial necessity or for other reasons work only part-time in the arts, will be allocated to some other occupation such as teacher or taxi driver. Hence the true number of people employed in a particular cultural occupation will be underestimated. On the other hand, since census forms are self-completed, some would-be or dilettante artists may declare themselves as full-time professionals when in fact their contribution to creative output in the economy is low or zero. Furthermore census data are seen to have particular problems in cities like London where transient populations, people with language difficulties, etc. are disproportionately represented (Higgs, Cunningham, & Bakhshi, 2008, pp. 34–35). Finally, the occupational descriptions as specified on census forms may not always correspond exactly to the detail of what creative people actually do.

To summarise, the empirical information required for any given country in order to test the hypothesis put forward above amounts to a two-way table of numbers of workers by industry and occupation; the table will specify individual cultural industries and the aggregate of non-cultural industries, whilst occupations will be disaggregated into creative and non-creative categories. Such a cross-tabulation has been used in earlier studies to examine the structure of the cultural workforce (Throsby, 2003), and has recently been examined more closely by Cunningham (2006) and Higgs et al. (2008), who interpret the two-by-two classification (creative/non-creative industries by creative/non-creative workers) as identifying a "creative trident", comprising three of the four quadrants in the table:

- *specialist workers*: those employed in core creative occupations within creative industries;
- *support workers*: those employed in other occupations within the creative industries; and
- *embedded workers*: those employed in core creative occupations within other industries.

These papers argue that the direct economic impact of the creative industries is frequently underestimated because the embedded workers are overlooked.

Application

The procedures outlined above were applied to data from five countries: Australia, Canada, New Zealand, the UK and the US. Detailed tabulations and data sources can be found in Throsby and Zednik (2007). This section of the present paper summarises the main results and tests the hypothesis put forward above concerning the structure of the concentric circles model.

The first problem encountered in data-gathering for this study was the fact that there is no common standard for identifying and classifying either cultural industries or cultural occupations across the countries under consideration that would enable a consistent allocation to be made of industries to categories or occupations to creative/non-creative status. Hence a number of approximations were necessary to accommodate the idiosyncrasies of the various countries' data sources.

In the case of the industry classification, each of the five countries uses different amalgamations of particular activities in their industry definitions for the cultural sector, making assignment of industries to the four categories delineated above occasionally problematical; for example, film is sometimes listed separately, and sometimes it is included with television. Furthermore, the number of individual cultural industries or groups of industries that could be identified varied

from country to country, from 10 in Canada to 24 in New Zealand. The cultural industries for which data could be obtained were assigned to one of the four categories according to the scheme outlined earlier in this paper; the specific industries included in each category for each country are summarised in Table 1 (fuller details are given in Throsby & Zednik, 2007).

In regard to occupational classifications, we were able to make use of the framework for cultural statistics developed by the Australian Bureau of Statistics (ABS) which has defined what it calls “cultural occupations” to include, in addition to the major category “artists and related professionals”, a number of other occupations such as architects, librarians, museum curators, technicians working in the arts, and so on (ABS, 1996). Adoption of this categorisation broadens the scope of cultural occupations considerably, and suggests that it is necessary to identify a subgroup of specifically *creative* occupations within the broader cultural occupation category, bearing in mind the conceptual basis of the concentric circles model.

Adopting the ABS cultural occupation categories and identifying a specifically creative subgroup within them led to the following classification:

Cultural occupations, comprising:

(a) *Creative occupations*

- Visual artists
- Photographers, sculptors, craftspeople
- Writers, editors
- Musicians, composers, singers
- Dancers, choreographers,
- Actors
- Directors

(b) *Other cultural occupations*

- Designers, architects
- Journalists, presenters
- Producers
- Librarians, curators, administrators
- Technicians
- Support personnel

Results

Employment data gathered for Australia, Canada, New Zealand and the UK for the year 2001 and for the US for 2004 are summarised in Appendix Tables 1–5, and the relevant percentages of creative and cultural employment to total employment in each of the concentric circles are shown in Appendix Table 6. Table 2 draws together the results required to test the hypothesis put forward above concerning the structure of the concentric circles model.

Given the assumptions of the model, it is apparent that the results in Table 2 are consistent with the hypothesis under review; i.e. they indicate that the proportion of creative labour used in production, as an indicator of the cultural content of the output of the industries included in the model, does indeed decline as one moves outwards from the centre of the concentric circles. This result holds true for all five countries and for both cultural employment and the more narrowly defined concept of creative employment. The rate of decline varies between countries because of differences in the capture of each country’s industries and occupations in each of

Table 1. Cultural industries included in each concentric-circles classification: Australia, Canada, New Zealand, UK and USA.

	Core creative arts	Other core creative industries	Wider cultural industries	Related industries
AUSTRALIA	Music and theatre production; Creative arts	Film and video publishing; Museums, libraries; Photographic studios; Services to the arts	Publishing/printing; Radio/TV; Film distribution and exhibition	Architecture, Advertising, Commercial Arts Display; Religious organisations; Parks and gardens; Other cultural industries
CANADA	Performing arts, spectator sports and related industries	Heritage institutions; Photographic services	Printing/publishing; Broadcasting and telecommunications; Motion picture and sound recording industries	Professional, scientific and technical services; Clothing, fashion, design; Religious, grant-making, civic, and professional and similar organisations
NEW ZEALAND	Music, theatre productions, performing arts venues; Creative arts	Film and video production; Museums, libraries; Photographic studios; Services to the arts not elsewhere defined	Printing/publishing; Radio/TV; Other film, video and sound	Advertising & commercial art; Architectural services; Religious organisations; Interest groups not elsewhere classified
UNITED KINGDOM	Artistic and literary creation & interpretation, live theatre, operation of arts facilities, dance halls, fair and amusement parks	Motion picture and video activities; Library; archives; museums and other cultural activities; Manufacture of musical instruments; Photographic services	Publishing/printing; Radio/TV; Miscellaneous business activities not elsewhere classified	Advertising; Architectural and engineering activities and related technical consultancy; Clothing/ fashion/design
UNITED STATES	Music and theatre production; Independent artists, writers, and performers	Motion picture and video industries; Museums, libraries; Photographic services; Services to the arts	Publishing/printing; Radio/TV; Sound recording industries	Advertising and related services; Architectural, engineering, and related services; Clothing, fashion, design; Religious organisations

Table 2. Proportions of creative/cultural employment to total employment in cultural industry groups: Australia (2001), Canada (2001), New Zealand (2001), UK (2001), USA (2004) (%).

Industry group	Australia	Canada	New Zealand	United Kingdom	United States
Proportion of total industry employment in creative occupations					
Core creative industries	65.3	46.7	53.7	41.7	42.5
Other core creative industries	17.5	25.7	17.5	13.7	21.2
Wider cultural industries	9.3	5.7	5.6	4.3	6.9
Related industries	1.4	0.8	2.1	1.7	2.1
Proportion of total industry employment in all cultural occupations					
Core creative industries	80.2	64.3	65.9	46.2	55.7
Other core creative industries	61.3	46.9	58.3	39.1	42.3
Wider cultural industries	46.1	22.1	21.1	25.5	20.6
Related industries	31.8	11.6	20.8	15.4	12.3

Source: Appendix Table 6.

the model's categories. For the same reason, the percentages cannot be compared across the table; it is not possible, for example, to conclude from Table 2 that core creative employment in Australia is necessarily higher in proportional terms than in the other countries shown.

These results may also be used to examine some wider features of the cultural workforce in the five countries, as shown in Table 3. For example, the total employment in the cultural industries as a proportion of total national employment levels is indicated in the first column of the table. In four of the five countries the cultural industries as defined in this study can be seen to account for around 4% of the workforce;¹⁰ the somewhat higher level in the UK arises because of the particular industries included in that country's data.¹¹

It was noted earlier that employment in the cultural industries alone may understate total cultural employment insofar as workers in cultural occupations employed in non-cultural industries are not counted. In the "creative trident" model, these workers are referred to as "embedded". Including these workers in total cultural employment raises the percentages as shown in the second column of Table 3. These figures suggest a range of total cultural employment from about 4% to almost 9% as a proportion of the total workforce in the countries under study.

Table 3. Characteristics of the cultural workforce: Australia, Canada, New Zealand, UK, USA.

	Cultural industry employment as proportion of total economy employment	Total cultural employment as proportion of total economy employment	"Embedded" employment as proportion of total employment in "specialist" occupations
	(%)	(%)	(%)
Australia (2001)	3.6	5.2	51.0
Canada (2001)	13.4 ^(a)	15.6 ^(a)	50.0
New Zealand (2001)	4.1	5.8	59.9
UK (2001)	7.5	8.7	40.0
USA (2004)	3.8	4.6	57.5

Source: Calculated from data in Appendix Tables 1–5.

Note: (a) These figures are anomalous because of the "Related Industries" group in the Canadian data. Omitting this group gives percentages of 4.0 and 4.3 respectively.

The significance of the embedded workers is highlighted in the last column of the table, which indicates that in most of the countries, the majority of cultural or creative workers work *outside* the cultural industries. Note that the figure for the UK can be compared with an estimate of 54% derived also for the year 2001 by Higgs et al. (2008, p. 4). This latter figure is based on somewhat different definitions of creative industries and occupations from those used above.

Conclusions

This paper has examined the assumptions and structure of the concentric circles model of the cultural industries. Empirical data from five countries are used to examine the model's key characteristic, the proposition that the cultural content of the output of the cultural industries declines as one moves outwards from the core. Since cultural content cannot be measured directly, the test is based on using the proportion of creative inputs (i.e. labour) employed in production as a proxy. Given the assumptions from which the model is derived, the results illustrate its validity as a means of depicting the structural characteristics of the cultural industries.

Of course there is no "right" or "wrong" model of the cultural industries, simply a range of alternative constructions based on different sets of assumptions and employing different mechanisms for putting the parts together. But the choice of a model by means of which to interpret the structural characteristics of cultural production has important implications for the formulation of cultural policy. For example, the estimated economic size of the cultural industries, measured in terms of value added, contribution to GDP, levels of employment etc., will vary considerably according to the model chosen, since different models include a different collection of industries in the mix.¹²

The appeal of the concentric circles model lies in its emphasis on primary creative ideas as the driving force that propels the cultural industries and that distinguishes them from other industries in the economy. This basic characteristic helps to link economic and cultural analysis of the creative sector. In particular the model's incorporation of the economic/cultural value distinction, and the centrality it attributes to the creative arts, may serve to strengthen the cultural orientation of cultural policy and to counteract a tendency towards interpreting cultural policy simply as an arm of economic policy and nothing more.

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Notes

1. Nevertheless a broader agenda for cultural policy has been around for some time in the field of development; see, for example, the Declaration arising from the World Conference on Cultural Policies held in Mexico City in July–August 1982 (UNESCO, 1982).
2. For an application of this model to Australian data, see Gibson, Murphy, and Freestone (2002).
3. For a detailed discussion of the definition of cultural goods, see McCain (2006) and Throsby (2006).
4. Including aesthetics, philosophy, anthropology, art history, etc.; see the collection of essays in Hutter and Throsby (2008).
5. As an illustration, the allocation of design to "Related Industries" could be contested. Some aspects of design could be seen as a creative activity eligible for inclusion in the core of the model, whilst others such as industrial design are essentially utilitarian rather than engaged in conveying cultural content. In practice, the overall design industry, which embraces a wide range of fields, would seem best

interpreted as being mainly involved with commercialisation, and hence is properly placed in the outer layers of the concentric circles.

6. Note that the production and distribution of some cultural goods involves many more distinct stages, and hence different industries, than this simple example suggests. For instance, the production and distribution of music involves the live performance industry, the music publishing industry, the recording industry, the broadcasting industry, etc., all of which are represented in the various layers of the concentric circles model.
7. Caves (2000) uses the term “humdrum” inputs to describe the latter type of worker.
8. See also Markusen, Wassall, DeNatale, and Cohen (2008) for an analysis of arts and cultural occupations in the US.
9. For a more detailed discussion see Throsby (2001b).
10. This figure and those in the next paragraph could be regarded as still broadly applicable at the present time, assuming no significant shifts in the years since those to which the data relate.
11. For example, inclusion of industry groups such as “Architectural and engineering activities and related technical consultancy” raises the UK total significantly.
12. For example, the WIPO model mentioned in the Introduction above includes in its most extensive version industries manufacturing hardware such as photocopiers and other equipment for distributing copyright material, leading to very large estimates of the size of the creative sector.

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Appendix

Appendix Tables 1–5 show the industry X occupation tabulations for the five countries included in this study. These tables are condensed from the more comprehensive data contained in Throsby and Zednik (2007), which also contains details of methodology, assumptions and sources. Note that in Appendix Tables 1–5 the symbol (*) indicates < 0.05 .

Appendix Table 6 shows for all five countries the percentages of employment in each occupational category to total employment for each layer of the concentric circles model. These proportions are used to test the hypothesis concerning the structure of the model, as shown in text Table 2.

Note that in all tables some totals do not sum exactly, owing to rounding errors.

Table 1. Employment in cultural occupations and industries: Australia: 2001 ('000).

Occupational group	Core creative industries	Other core cultural industries	Wider cultural industries	Related industries	Total cultural industries	Total non-cultural industries	Total employment
Visual artists, craftspeople, etc.	4.0	3.8	1.5	0.8	10.1	11.9	22.0
Writers, editors	1.6	0.2	4.2	0.8	6.7	2.1	8.9
Musicians, composers, singers	6.0	0.3	0.1	0.2	6.6	10.8	17.4
Dancers, choreographers	0.2	0.1	*	0.1	0.4	4.4	4.8
Actors	1.2	0.4	0.6	0.2	2.4	2.9	5.3
Directors	0.2	1.4	1.2	0.2	3.0	0.8	3.9
Sub-total creative occupations	13.2	6.2	7.6	2.3	29.3	33.0	62.2
Designers, architects	0.6	0.8	2.9	31.1	35.4	30.7	66.1
Journalists, presenters	1.2	0.1	9.0	0.4	10.7	2.5	13.2
Producers	0.3	1.3	3.5	1.4	6.5	0.9	7.3
Librarians, curators, administrators	0.3	7.0	3.4	14.1	24.7	18.9	43.5
Technicians	0.4	3.8	7.0	1.4	12.6	34.5	47.1
Support personnel	0.2	2.9	4.2	0.8	8.2	12.2	20.4
Sub-total other cultural occupations	3.0	15.9	29.9	49.2	98.0	99.6	197.6
Total cultural occupations	16.2	22.1	37.5	51.4	127.3	132.6	259.8
Total non-cultural occupations	3.9	13.9	43.9	110.2	171.9	7,866.8	8,038.7
Total employment	20.2	36.0	81.4	161.6	299.2	7,999.3	8,298.5

Source: Calculated from data in Australian Bureau of Statistics (2001), *Employment in Culture, Australia* (Cat. no. 6273.0).

Table 2. Employment in cultural occupations and industries: Canada: 2001 ('000).

Occupational group	Core creative industries	Other core cultural industries	Wider cultural industries	Related industries	Total cultural industries	Total non-cultural industries	Total employment
Visual artists, craftspeople, etc.	6.1	4.6	1.1	3.0	14.7	8.7	23.5
Writers, editors	3.0	*	7.4	3.3	13.7	4.0	17.7
Musicians, composers, singers	3.4	*	0.3	0.4	4.1	2.5	6.6
Dancers, choreographers	0.2	*	*	*	0.2	0.9	1.2
Actors	1.5	*	0.2	*	1.8	0.5	2.3
Directors	1.4	*	7.6	0.6	9.6	0.8	10.4
Sub-total creative occupations	15.7	4.6	16.7	7.3	44.2	17.4	61.7
Designers, architects	0.5	0.1	8.2	40.3	49.2	28.5	77.6
Journalists, presenters	1.8	0.3	10.6	5.5	18.2	10.0	28.3
Librarians, curators, administrators	2.0	2.1	17.9	24.9	46.9	104.4	151.3
Technicians	1.1	1.2	9.0	13.0	24.4	18.7	43.2
Support personnel	0.4	0.1	2.4	3.5	6.4	10.4	16.7
Sub-total other cultural occupations	6.0	3.8	48.1	87.2	145.1	172.0	317.1
Total cultural occupations	21.6	8.4	64.8	94.5	189.4	189.4	378.8
Total non-cultural occupations	12.0	9.4	228.8	722.2	972.5	7,334.0	8,306.4
Total employment	33.6	17.9	293.7	816.8	1,161.8	7,523.4	8,685.2

Source: Calculated from data in Statistics Canada (2001), *2001 Census, Canada's Workforce: Paid Work*. (Cat. no. 9F0012XCB2001049).

Table 3. Employment in cultural occupations and industries: New Zealand: 2001 ('000).

Occupational group	Core creative industries	Other core cultural industries	Wider cultural industries	Related industries	Total cultural industries	Total non-cultural industries	Total employment
Visual artists, craftspeople, etc.	0.6	0.9	0.2	0.2	1.9	2.2	4.1
Writers, editors	0.4	0.1	1.0	0.2	1.7	1.2	2.9
Musicians, composers, singers	0.6	*	*	0.1	0.8	3.0	3.8
Dancers, choreographers	0.1	*	*	*	0.1	0.6	0.7
Actors	0.4	0.3	0.1	*	0.8	0.6	1.4
Directors	0.1	0.4	0.2	0.1	0.8	0.2	1.0
Sub-total creative occupations	2.2	1.8	1.5	0.6	6.1	7.8	13.9
Designers, architects	0.1	0.2	0.5	4.5	5.4	5.4	10.7
Journalists, presenters	0.2	0.1	2.5	0.1	2.8	0.8	3.6
Librarians, curators, administrators	0.1	3.2	0.6	0.2	4.1	5.0	9.1
Technicians	*	0.1	0.1	*	0.2	0.3	0.5
Support personnel	*	0.6	0.6	0.6	1.8	11.0	12.8
Sub-total other cultural occupations	0.4	4.2	4.2	5.4	14.3	22.4	36.7
Total cultural occupations	2.7	6.0	5.7	6.0	20.4	30.3	50.6
Total non-cultural occupations	1.5	4.4	21.4	22.9	49.9	1,626.7	1,676.6
Total employment	4.1	10.3	27.0	28.9	70.3	1,657.0	1,727.2

Source: Calculated from data in Statistics New Zealand (2001), *2001 Census of Population and Dwellings (MB2001)*, commissioned table.

Table 4. Employment in cultural occupations and industries: UK: 2001 ('000).

Occupational group	Core creative industries	Other core cultural industries	Wider cultural industries	Related industries	Total cultural industries	Total non-cultural industries	Total employment
Visual artists, craftspeople, etc.	8.9	5.0	28.4	6.3	48.5	87.3	135.8
Writers	7.4	0.7	14.1	2.8	25.0	19.2	44.1
Musicians, composers, singers	14.2	0.3	1.7	0.5	16.7	10.2	26.9
Dancers, choreographers	2.2	*	0.4	0.2	2.8	4.2	7.0
Actors	19.8	0.7	3.1	0.6	24.2	8.9	33.1
Directors, producers	5.6	2.6	4.6	3.2	16.1	5.6	21.7
Sub-total creative occupations	58.0	23.2	38.4	13.6	133.3	135.3	268.6
Designers, architects	2.7	1.1	38.0	79.7	121.4	72.0	193.4
Journalists, presenters, editors	2.3	3.0	82.8	3.8	91.9	45.2	137.1
Librarians, curators, administrators	0.8	35.4	6.3	22.0	64.4	45.7	110.2
Technicians	*	1.8	*	*	1.9	1.2	3.2
Support personnel	0.4	1.6	64.3	2.5	68.8	21.6	90.4
Sub-total other cultural occupations	6.2	42.9	191.4	107.9	348.4	185.8	534.2
Total cultural occupations	64.2	66.2	229.8	121.6	481.7	321.1	802.9
Total non-cultural occupations	74.8	103.2	669.9	665.6	1,513.5	24,259.4	25,772.9
Total employment	139.0	169.3	899.7	787.2	1,995.3	24,580.5	26,575.8

Source: Calculated from data in Office for National Statistics (2001), *2001 Census*, Table C0417, Crown Copyright 2004.

Table 5. Employment in cultural occupations and industries: USA: 2004 ('000).

Occupational group	Core creative industries	Other core cultural industries	Wider cultural industries	Related industries	Total cultural industries	Total non-cultural industries	Total employment
Visual artists, craftspeople, etc.	4.9	44.6	11.0	10.0	70.4	32.9	103.4
Writers	2.4	5.7	75.1	11.1	94.3	45.8	140.1
Musicians, composers, singers	32.9	2.2	0.4	10.4	45.9	15.9	61.8
Dancers, choreographers	5.8	0.4	*	*	6.1	25.7	31.8
Actors	17.3	80.8	0.2	*	98.3	29.4	127.7
Directors, producers	7.3	18.5	28.5	16.4	70.7	15.8	86.5
Sub-total creative occupations	70.5	152.2	115.2	47.9	385.8	165.6	551.4
Designers, architects	3.4	6.9	44.8	184.4	239.5	220.2	459.7
Journalists, presenters, editors	6.4	9.7	103.9	31.9	151.8	178.2	330.0
Librarians, curators, administrators	5.9	77.8	7.3	8.3	99.2	223.8	323.1
Technicians	4.3	27.4	35.1	5.5	72.2	177.5	249.7
Support personnel	1.9	29.5	39.6	3.1	74.1	165.9	240.0
Sub-total other cultural occupations	21.9	151.4	230.6	233.1	636.9	965.6	1,602.6
Total cultural occupations	92.4	303.6	345.8	281.0	1,022.7	1,131.2	2,153.9
Total non-cultural occupations	73.5	414.2	1,332.1	2,002.9	3,822.6	123,170.2	126,992.8
Total employment	165.9	717.7	1,677.9	2,283.9	4,845.4	124,301.4	129,146.8

Source: Calculated from data in Bureau of Labor Statistics (2004), *November 2004 National Industry-Specific Occupational Employment and Wages Estimates*.

Table 6. Employment in creative/cultural occupations as a percentage of total employment in cultural industry groups: Australia (2001), Canada (2001), New Zealand (2001), UK (2001), USA (2004) (%).

	Core creative industries	Other core cultural industries	Wider cultural industries	Related industries	Total cultural industries	Total non-cultural industries	Total employment
AUSTRALIA							
Creative occupations	65.3	17.5	9.3	1.4	9.8	0.4	0.7
Other cultural occupations	14.8	43.8	36.7	30.4	32.8	1.2	2.4
Total cultural	80.2	61.3	46.1	31.8	42.5	1.7	3.1
Non-cultural	19.8	38.6	53.9	68.2	57.5	98.3	96.9
Total employment	100.0	100.0	100.0	100.0	100.0	100.0	100.0
CANADA							
Creative occupations	46.4	25.7	5.7	0.8	3.8	0.2	0.7
Other cultural occupations	17.9	21.2	16.4	10.7	12.5	2.3	3.7
Total cultural	64.3	46.9	22.1	11.6	16.3	2.5	4.4
Non-cultural	35.7	53.1	77.9	88.4	83.7	97.5	95.6
Total employment	100.0	100.0	100.0	100.0	100.0	100.0	100.0
NEW ZEALAND							
Creative occupations	56.1	17.5	5.6	2.1	8.7	0.5	0.8
Other cultural occupations	9.8	40.8	15.6	18.7	20.3	1.4	2.1
Total cultural	65.9	58.3	21.1	20.8	29.0	1.8	2.9
Non-cultural	34.1	41.7	78.9	79.2	71.0	98.2	97.1
Total employment	100.0	100.0	100.0	100.0	100.0	100.0	100.0
UK							
Creative occupations	41.7	13.7	4.3	1.7	6.7	0.6	1.0
Other cultural occupations	4.5	25.3	21.3	13.7	17.5	0.8	2.0
Total cultural	46.2	39.1	25.5	15.4	24.1	1.3	3.0
Non-cultural	53.8	60.9	74.5	84.6	75.9	98.7	97.0
Total employment	100.0	100.0	100.0	100.0	100.0	100.0	100.0
USA							
Creative occupations	42.5	21.2	6.9	2.1	8.0	0.1	0.4
Other cultural occupations	13.2	21.1	13.7	10.2	13.1	0.8	1.2
Total cultural	55.7	42.3	20.6	12.3	21.1	0.9	1.7
Non-cultural	44.3	57.7	79.4	87.7	78.9	99.1	98.3
Total employment	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Source: Calculated from data in Appendix Tables 1–5.